

TSI Sign: Product Overview

The open-source, self-hosted document engine that gives you full data sovereignty.

Executive Summary TSI Sign is an end-to-end, self-hosted document generation, execution, and signing engine designed for organizations that demand absolute control over their data infrastructure. Built on open-source principles, TSI Sign eliminates third-party SaaS platform markups and data routing, allowing enterprises to handle dynamic template rendering, document workflows, digital seals, and legally binding identities entirely on-premise or within their private cloud.

Core Architecture & Capabilities

- **Self-Hosted Data Sovereignty:** Run the engine inside your own perimeter (Docker/Kubernetes). Templates, payloads, documents, and audit logs never touch an external vendor's cloud storage.
- **Integrated Document Generation & Template Engine:**
 - *HTML5/CSS3 Template Management:* Store versioned document templates with dynamic placeholder tags (`{{variable}}`) and CSS-based visual signature anchors.
 - *High-Performance PDF Compilation:* Native Java-based HTML-to-PDF compilation via **OpenHTMLtoPDF** (backed by **Apache PDFBox**), generating production-ready PDF/A files directly from dynamic JSON payloads.
- **Dual-Mode Signing Engine:**
 - *Cryptographic PKI Seals:* Secure internal workflows, system-generated invoices, and corporate records locally using enterprise Java **KeyStore** (`.pfx/.p12`) certificates or hardware tokens (PAdES standard).
 - *Identity-Backed External Signatures:* Connect directly to licensed Certifying Authority (CA) gateways using your organization's own API credentials for compliant remote signing.
- **Granular Audit Trails:** Automatically compiles robust cryptographic audit logs containing timestamping, session metadata, IP tracking, and tamper-evident document hashing.
- **Zero SaaS Markup:** Pay statutory fees or direct CA costs at raw market rates with zero intermediary transaction fees or monthly seat licensing penalties.

Ideal Use Cases

- **Internal Corporate Governance:** Automated pre-filling and sealing of board resolutions, HR contracts, and internal policy approvals.
- **Regulated Industries (BFSI & Healthcare):** Localized document generation and processing for organizations restricted by strict data-residency laws.
- **High-Volume Enterprise Operations:** Deploying an end-to-end document engine

microservice integrated seamlessly into existing ERP, CRM, or document management systems (DMS).

The unSaaS Advantage TSI Sign flips traditional document software on its head. Instead of renting workflow capacity or paying separate fees for template generation and signature execution from closed cloud vendors, deploying organizations own their infrastructure, configure their own CA relationships, and retain permanent ownership of their document ecosystem.

Several open-source eSignature products exist, though **none are natively tailored for India’s specific legal and statutory stack** (Aadhaar eSign, NeSL, SHCIL eStamping) like TSI Sign aims to be.

The global open-source landscape includes:

- **Documenso:** Often dubbed "the open-source DocuSign". Built with Next.js, it focuses heavily on modern UI, clean developer APIs, and embedding signing flows directly into web apps.
- **DocuSeal:** A lightweight, highly popular self-hosted tool optimized for automated document workflows, HTML-to-PDF fields, and frictionless form signing.
- **OpenSign:** An open-source document signing solution focusing on basic PDF field placement, multi-signer routing, and email OTP verification.
- **LibreSign:** A self-hosted electronic signature tool tightly integrated with the Nextcloud ecosystem, popular among organizations looking for self-hosted productivity suites.

How TSI Sign Stands Out

Global open-source tools handle basic web-drawing signatures and generic PDF cryptographic hashes well, but they lack out-of-the-box integration with Indian regulatory infrastructure.

TSI Sign bridges this gap by acting as a **self-hosted, India-compliant unSaaS engine**:

Feature / Capability	Global Open-Source Tools (Documenso, DocuSeal)	TSI Sign
Local PKI / PDF Sealing	Supported (standard PAdES/RSA)	Supported (PAdES with local <code>.pfx</code> or HSM)

Aadhaar OTP eSign Gateway	Not available (requires custom dev)	Native Adapter (direct eMudhra/C-DAC API)
Indian Legal Validity	Relies on generic electronic consent	Built for IT Act Section 3A compliance
Data Sovereignty	High (Self-Hosted)	High (Self-Hosted + Direct Statutory Billing)

TSI Sign fills the exact niche for Indian enterprises wanting a self-hosted platform without paying SaaS markups or custom-building eSign API connectors from scratch.
